

The first antisense in the clinic

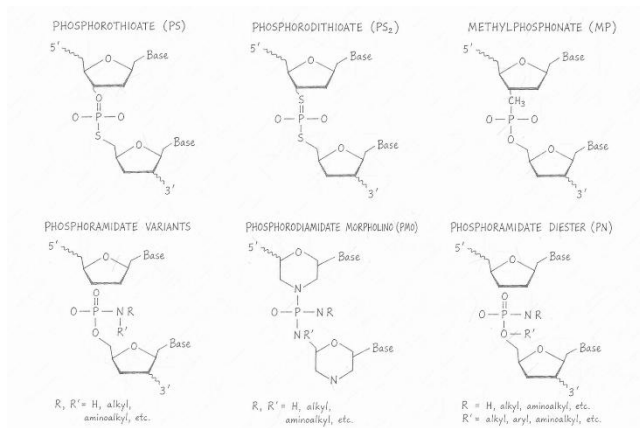


FOMIVIRSEN (BRAND NAME VITRAVENE)

Fomivirsen is a simple sequence specific antisense DNA molecule with a modified phosphorothioate backbone (PS); this backbone is the only thing protecting it from nuclease activity.

Intravitreal administration of Fomivirsen significantly prolonged the median time to disease progression in patients with CMV retinitis compared to delayed treatment, supporting its therapeutic efficacy (Geary, Henry and Grillone, 2002). Fomivirsen demonstrated activity against CMV strains resistant to conventional antivirals and exhibits additive effects when used in combination with agents such as ganciclovir or foscarnet (Azad et al., 1995).

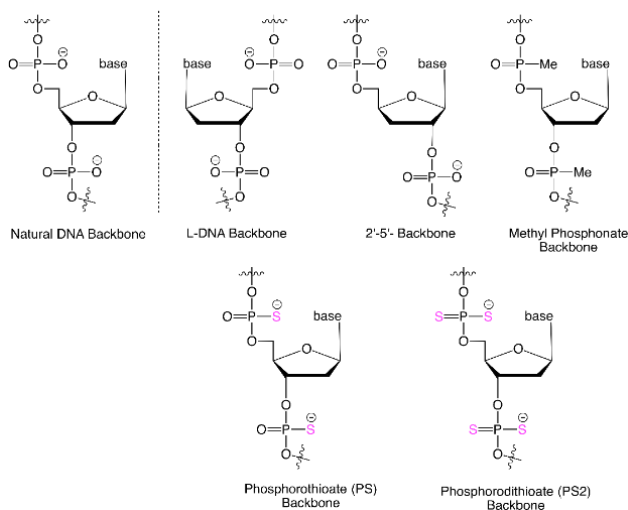
Charge, Stability and Function



Chemical modification of the oligonucleotide backbone underpins the success of modern nucleic acid therapeutics. While native phosphodiester (PO) linkages provide optimal compatibility with biological machinery, they are intrinsically unstable and pharmacologically limited. A diverse set of backbone chemistries including phosphorothioate (PS), phosphorodithioate (PS₂), methylphosphonate (MP), phosphoramidate variants, phosphorodiamidate morpholino (PMO), and emerging phosphoramidate diester (PN) chemistries has emerged to address these constraints. These modifications operate within a constrained design space

defined by three interdependent variables: charge, nuclease stability, and biological mechanism compatibility.

First, electrostatic charge governs interactions with proteins, membranes, and biological barriers. The phosphodiester backbone (PO) is fully anionic, enabling strong protein binding and efficient recognition by endogenous enzymes, but simultaneously limiting membrane permeability and promoting nonspecific interactions. Phosphorothioate (PS) linkages retain this negative charge while altering its distribution, contributing to increased protein binding and improved pharmacokinetics. In contrast, several alternative chemistries including methylphosphonates, phosphoramidates, phosphorodiamidate morpholino (PMO), and PN linkages reduce or eliminate backbone charge. PMO and PN backbones are largely charge-neutral, fundamentally altering electrostatic interactions, often reducing plasma protein binding and accelerating clearance while potentially improving intracellular distribution. However, reduced charge can also diminish hybridization affinity and weaken protein-mediated uptake pathways that are critical for productive delivery (Crooke, 2017; Juliano, 2016).



Second, nuclease stability is primarily governed by chemical substitutions at the phosphorus center and by steric protection from sugar modifications. Substitution of non-bridging oxygen atoms with sulfur (e.g., PS linkages) reduces nuclease-mediated hydrolysis and increases resistance to degradation in biological matrices (Lovric et al., 2025) (Wan et al., 2016). Importantly, stability is not solely dependent on the presence of stabilizing chemistries but also on their distribution and combination. Mixed backbone designs (e.g., PO/PS or PS/PN chimeras) and gapmer architectures allow selective tuning of degradation kinetics and pharmacokinetics. For example,

incorporation of PO linkages within a PS backbone reduces stability in a position-dependent manner, while sugar modifications such as MOE can compensate for reduced backbone protection and enable fine-tuning of tissue distribution (Lovrić et al., 2025; Geary, 2001).

The following table highlights typical rules associated with the placement of PO linkages in a gapmer ASOs:

Phosphodiester (PO) placement

At very 5' end	Often increases nuclease vulnerability, especially to exonucleases. May reduce protein binding and alter uptake.
At very 3' end	Often risky because many nucleases attack from the 3' end. A terminal or near-terminal PO can create a cleavage-sensitive site.
Near ends but not terminal	Can reduce stability, but effect depends on neighboring bases and sugar modifications. Sometimes tolerated if protected by 2'-modified sugars such as MOE, cEt, LNA, or 2'-OMe.
Central DNA gap of RNase H gapmer	Usually the most dangerous place for PO if nuclease exposure is high, because the DNA gap is already less sugar-protected than MOE wings. PO in the gap can reduce metabolic stability. It may also affect RNase H activity depending on placement.
In MOE / 2'-modified wings	More tolerated than in an unmodified DNA gap, because sugar modifications add nuclease resistance. PO here may be used to tune protein binding, clearance, or toxicity.
Multiple PO linkages clustered together	Usually lowers stability more strongly. Lovrić et al. reported that ASOs with one PO were more stable than those with two or three PO links in nuclease-containing systems.
PO next to MeC	Lovrić et al. found that the local sequence context matters; their study reports that PO placement relative to 5-methylcytidine (see modified nucleobases section) affected nuclease resistance, meaning "PO before" and "PO after" the same base are not necessarily equivalent.

ASOs in the clinic have two main backbone strategies: fully phosphorothioated designs and newer mixed PS/PO gapmers. Fomivirsen is a first-generation 21-mer DNA ASO with a fully phosphorothioate backbone, meaning all 20 internucleotide linkages are PS and there are no PO linkages (Anderson et al., 1998). Several second-generation 5-10-5 MOE/DNA gapmers follow the same all-PS backbone logic: mipomersen,

inotersen and volanesorsen are each 20-mer ASOs with all PS and no PO linkages, with five 2' MOE residues at each end and a central ten-nucleotide DNA gap (Geary et al., 2015; Ionis Pharmaceuticals, 2013; Ionis Pharmaceuticals, 2019; Witztum et al., 2019). Nusinersen is different mechanistically because it is a splice-modulating ASO rather than an RNase H gapmer, but it is also fully phosphorothioated: as an 18-mer 2'-MOE ASO, it contains 17 PS linkages and 0 PO linkages (Bennett et al., 2017). By contrast, tofersen uses an explicitly mixed PS/PO 5-10-5 MOE gapmer backbone: it is a 20-mer with 19 total internucleotide linkages, of which 15 are PS and 4 are PO; in 5'→3' notation its linkage pattern is PS–PO–PS–PO–PS–PS–PS–PS–PS–PS–PS–PS–PS–PS–PO–PS–PO–PS, placing the four PO linkages in the MOE wing regions rather than across the central DNA gap (Biogen, 2023; KEGG, 2026). This pattern preserves an all-PS DNA gap for RNase H1-competent activity while reducing total PS burden at selected wing positions.

Chemistry	Pattern	Charge	Used by	In approved drugs?
PO	soooooooooooooos	full negative	Alnylam	Onpattro and others
PS	ssssssssssssssss	full negative	Ionis	Yes, many including Spinraza
PS + PO mix	sooooo:sssssssssooss	mixed	n-Lorem	Yes, neuro – gapmer
PN	ssssnssssssssnsss	reduced	Wave	
Stereopure PS		full negative	Wave	

New backbone chemistries are being developed to retain the useful pharmacology of PS ASOs while reducing liabilities associated with extensive PS content, including nonspecific protein binding, pro-inflammatory effects and sequence-dependent toxicity. One prominent example is mesyl phosphoramidate chemistry, also referred to as MsPA or μ chemistry, in which a mesyl phosphoramidate group replaces the natural phosphodiester linkage at selected positions. Unlike many neutral backbone analogues, MsPA remains negatively charged near physiological pH and acts as a close phosphodiester mimic, while providing nuclease resistance that can exceed PS and preserving compatibility with RNase H recruitment in appropriate DNA-like ASO designs (Stetsenko, 2023). This makes MsPA particularly attractive for next-generation gapmers, where mixed PS/MsPA backbones can be used to preserve RNase H1 activity and productive cellular uptake while reducing the total PS burden. In Ionis’ next-generation ASO work, site-specific replacement of PS linkages with MsPA in gapmer ASOs improved nuclease stability, reduced nonspecific protein binding, lowered cytotoxic and pro-inflammatory effects, and extended pharmacodynamic duration in vivo; importantly, replacement of selected PS linkages, including several within the DNA gap, was tolerated without loss of RNase H1-mediated activity, whereas complete replacement of all PS